When $y = 0$, (2) gives $x = 2$.

Thus $Q(2,0)$ is another point on the line.

Plot P and Q; join them to produce the line (2).

The point of intersection (3, 2) of lines (1) and (2) is a solution.

The solution is the point that is common to both the lines. Here we find it to be (3,2). We can give the solution as $x = 3$ and $y = 2$.

Note

It is always good to verify if the answer obtained is correct and satisfies both the given equations.

Example 3.45

Use graphical method to solve the following system of equations: $3x + 2y = 6$; $6x + 4y = 8$

Solution

Let us form table of values for each line and then fix the ordered pairs to be plotted.

Graph of $3x + 2y = 6$

x	−2	0	2
y	6	3	0

Points to be plotted :

(−2,6), (0,3), (2,0)

Graph of $6x + 4y = 8$

x	−2	0	2
y	5	2	−1

Points to be plotted :

(−2,5), (0,2), (2,−1)

When we draw the graphs of these two equations, we find that they are parallel and they fail to meet to give a point of intersection. As a result there is no ordered pair that can be common to both the equations. In this case there is no solution to the system.

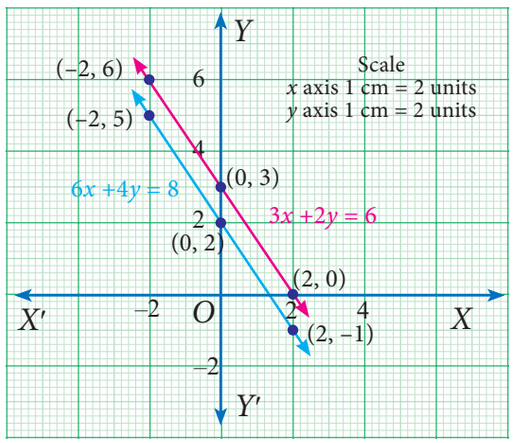


Fig. 3.22

Example 3.46

Use graphical method to solve the following system of equations: $y = 2x + 1$; $-4x + 2y = 2$

Solution

Let us form table of values for each line and then fix the ordered pairs to be plotted.

Graph of $y = 2x + 1$

Graph of $-4x + 2y = 2$

$2y = 4x + 2$

$y = 2x + 1$

3 Algebra_Term1.indd 122 02/12/2022 21:40:13